Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/14/2026
PI Name:	Allaway, Heather Ph.D.		
Project Title:	A Time Course of Bone Microarchitectural and Material Property Changes in Male and Female Mice During Simulated Unloading and Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70803-0001	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity
Start Date:	03/01/2024	**End Date:**	02/28/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/2026 per S. Mack-Phillips/NASA JSC (Ed., 2/12/2026).		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Lau, Anthony Ph.D. (The College of New Jersey)		
Grant/Contract No.:	80NSSC24K0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The evaluation of sex differences in the time course of physiologic responses and adaptations to the space environment is critical to ensuring the safety of crews as the world prepares for the first crewed spaceflights beyond low Earth orbit in 50 years and the first woman to step on the Moon. Microgravity during spaceflight damages musculoskeletal tissues, but the understanding of sex differences in musculoskeletal atrophy during spaceflight is incomplete. Maintaining musculoskeletal health during long-duration spaceflight is crucial for ensuring astronauts are physically ready for exploration class missions and mission success upon return to partial weightbearing activity. Due to the known links between sex and musculoskeletal health, it is critical to get a better understanding of “how” and “why” bone microarchitectural and material property changes occur over a mission duration and if that change occurs differently in males and females. Finite element (FE) modeling protocols and computational models have been developed to better identify how bone strength is compromised during reduced weight-bearing and other challenges. Preliminary data using FE modeling on female mice flow on Space Shuttle mission STS-135 indicated that proximal tibia bone density loss (-17%) underpredicted the attenuation in structural strength (-34%). These findings demonstrate a limitation in the current clinical assessments of bone health and fracture risk using dual energy x-ray absorptiometry (DXA), as well as the standard micro computed tomography (microCT) analysis for predicting bone health during spaceflight. The objective of the current proposal is to capitalize on samples from two completed projects that the team has already acquired through tissue sharing opportunities. The parent projects had male and/or female mice that underwent unloading (hindlimb unloading or spaceflight) for 14, 28, or 38 days (and appropriate controls). We will quantify bone cellular, microarchitecture, and strength changes using histology, microCT, computational FE modeling, and material property characterization. Successful completion of this project will provide foundational data for further mechanistic and translational studies examining sex differences in the musculoskeletal adaptation to spaceflight environmental stressors that will enable more specific countermeasure development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The work is significant to NASA’s goal for maximizing astronaut health during long-duration spaceflight. The proposed work will provide critical insight on how changes in bone microarchitecture and material properties during spaceflight contribute to overall bone strength, which are important factors for understanding and mitigating fracture risk. The proposed work will synergize with the development of novel imaging modalities to monitor bone health and countermeasure development, which will benefit astronauts and people on Earth.
Task Progress:	The proposed work uses a multi-scale approach to assess changes in bone health and strength in already available hindlimb bone specimens obtained from tissue sharing opportunities: 1) male mice from a 38-day International Space Station (ISS) flight study (80NSSC19K0392) and 2) male and female mice from a HU time course (14 vs 28 days) study (80NSSC19K1025). All live animal work was completed by July 2022 under approved Institutional Animal Care and Use Committee (IACUC) protocols at the parent study institutions. In the past reporting period, we have made progress toward the completion of SA1 and SA2, but have been delayed due to equipment issues and construction causing data variability issues. SA1: Determine the time course of bone microarchitectural and material property changes from long-duration unloading and the subsequent impact on bone strength. Calcified and decalcified bone histology: Graduate student training on microtoming, staining, and histomorphological quantification using the BioQuant software is underway. Student training on microtoming is going slowly as we are being extremely cautious to not lose any precious mouse tissue samples due to a lack of skill confidence. Training on the Bioquant system has progressed substantially once the computer for the system was replaced. Consistency in the identification of cells for the analyses is becoming better. Imaging on the microscope has been difficult due to vibration artifacts that develop due to construction happening in the building. When viewed, these images are retaken. All samples have been decalcified and paraffin-embedded. 1/3 of the embedded in plastic resin has been sectioned and stained. This was delayed by damage caused to the microtome, and the manufacturer of the microtome delaying the repairs. The remaining samples must be reembedded due to issues with the angle of the sample, preventing the section that would be cut from being in the correct position for analysis. Reembedding was delayed due to the discontinuation of one of the chemicals needed for the solution used for the embedding process. Ex vivo microCT Imaging & Analysis: uCT scanning must be redone after being delayed due to construction in the microCT building, resulting in vibration artifacts. Rescanning will not be performed in Dr. Lau’s lab, enabling biomechanical testing to occur prior to rescanning. Finite Element (FE) Modeling: Nothing to report. Cannot begin FEA analysis until uCT scans are complete. Biomechanical Testing: 3-point-bend-to-failure and spherical micro-indentation tests are complete. Statistical analyses have not been performed to keep the research team blinded while sample analyses continue. SA2: Evaluate sex differences in the time course of bone microarchitectural, material property, and strength changes from long-duration unloading. During the past reporting period, all researchers have remained blinded to the sex of the animals. Data collection has not progressed far enough to unblind the research team and evaluate this question. Progress has been made on data collection, as all the data for SA1 is used in the evaluation of SA2.
Bibliography Type:	Description: (Last Updated: 01/28/2026)
Abstracts for Journals and Proceedings	Beard C, Lau AG, Bowman LC, Allaway HCM. "Methodological approaches to assess sex differences in the effects of hindlimb unloading on bone strength in mice." 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025